

Module 2 Lesson 4

Applications of Linear Equations (6.3)

Objectives

In this lesson, we will learn how to

- use linear equations to solve problems;
- solve a formula for a variable.

1. Strategy for Solving Word Problems

Word problems can often be solved by translating English phrases into algebraic equations.

Strategy for Solving Word Problems

1. Read the problem carefully several times until you can state in your own words what is given and what the problem is asking for. Let x , or another variable, represent one of the unknown quantities.
2. If necessary, write expressions for any other unknown quantities in terms of x .
3. Write an equation in x that models the conditions of the problem.
4. Solve the equation and answer the question.
5. Check the solution in the original wording of the problem, not only in the equation.

2. Algebraic Translations of English Phrases

Many word problems use common phrases that suggest mathematical operations.

Common Algebraic Translations

Operation	Common Phrases
Addition	sum, more than, increased by
Subtraction	minus, decreased by, subtracted from, difference between, less than, fewer than
Multiplication	times, product of, percent of a number, multiplied by, twice
Division	divided by, quotient, reciprocal

3. Application: Education Pays Off

The median yearly after-tax income of a full-time worker with a bachelor's degree exceeds that of a full-time worker with an associate's degree by \$11 thousand.

The median yearly after-tax income of a full-time worker with a master's degree exceeds that of a full-time worker with an associate's degree by \$21 thousand.

Combined, three full-time workers with each of these degrees earn \$149 thousand after taxes.

Example 1: Find the median yearly after-tax income of full-time workers with each of these levels of education:

- associate's degree;
- bachelor's degree;
- master's degree.

Solution

Let

x = the median yearly after-tax income of a worker with an associate's degree.

Then

$x+11$ = the median yearly after-tax income of a worker with a bachelor's degree,

and

$x+21$ = the median yearly after-tax income of a worker with a master's degree.

Write the equation:

$$x + (x + 11) + (x + 21) = 149.$$

Combine like terms:

$$3x + 32 = 149.$$

Subtract 32 from both sides:

$$3x = 117.$$

Divide by 3:

$$x = 39.$$

Thus,

associate's degree: \$39,000,

bachelor's degree: $39 + 11 = 50$, or \$50,000,

master's degree: $39 + 21 = 60$, or \$60,000.

4. Application: Modeling Options for a Toll

While planning a Florida vacation, you find out that you need to cross a toll bridge.

One payment option requires purchasing a transponder for \$20. With the transponder, you pay a reduced toll of \$3.25 each time you cross the bridge.

Another option is toll-by-plate. With this option, you pay the full toll of \$4.25 each time you cross the bridge, plus a \$3 administrative fee for each crossing.

Example 2: Find the number of times you would need to cross the bridge for the costs of the two options to be the same.

Solution

Let

x = the number of times you cross the bridge.

The cost with the transponder is

$$20 + 3.25x.$$

The toll-by-plate cost is

$$(4.25 + 3)x = 7.25x.$$

Set the two costs equal:

$$7.25x = 20 + 3.25x.$$

Subtract $3.25x$ from both sides:

$$4x = 20.$$

Divide by 4:

$$x = 5.$$

The costs are the same after

$$5$$

crossings.

Check:

$$\text{Toll-by-plate} = 4.25(5) + 3(5) = 21.25 + 15 = 36.25.$$

$$\text{Transponder} = 20 + 3.25(5) = 20 + 16.25 = 36.25.$$

Both options cost \$36.25.

5. Application: A Price Reduction

Your favorite electronics retailer is having a sale on noise-canceling wireless headphones.

After a 40% price reduction, you purchase the headphones for \$192.

Example 3: What was the price of the headphones before the reduction?

Solution

Let

x = the original price of the headphones.

A 40% reduction means the sale price is

$$x - 0.40x.$$

Write the equation:

$$x - 0.40x = 192.$$

Combine like terms:

$$0.60x = 192.$$

Divide by 0.60:

$$x = 320.$$

The original price was

$$\$320.$$

Check:

$$320 - 0.40(320) = 320 - 128 = 192.$$

6. Solving a Formula for a Variable

Sometimes we need to solve a formula for a different variable.

This means rewriting the formula so that the chosen variable is isolated.

Solving a Formula

To solve a formula for a variable:

1. Treat the formula like an equation.
2. Use inverse operations to isolate the desired variable.
3. Keep the equation balanced by doing the same operation to both sides.

7. Solving a Formula for l

The perimeter of a rectangle is

$$P = 2l + 2w,$$

where P is perimeter, l is length, and w is width.

Example 4: Solve the formula

$$P = 2l + 2w$$

for l .

Solution

Start with

$$P = 2l + 2w.$$

Subtract $2w$ from both sides:

$$P - 2w = 2l.$$

Divide both sides by 2:

$$\frac{P - 2w}{2} = l.$$

Thus,

$$l = \frac{P - 2w}{2}.$$

This may also be written as

$$l = \frac{P}{2} - w.$$

8. Solving a Formula for p

The total price of an article purchased on a monthly deferred payment plan is described by

$$T = D + pm,$$

where

$$T = \text{total price},$$

$$D = \text{down payment},$$

$$p = \text{monthly payment},$$

and

m = number of months.

Example 5: Solve the formula

$$T = D + pm$$

for p .

Solution

Start with

$$T = D + pm.$$

Subtract D from both sides:

$$T - D = pm.$$

Divide both sides by m :

$$\frac{T - D}{m} = p.$$

Thus,

$$p = \frac{T - D}{m}.$$

9. More Practice

Example 6: A gym charges a one-time registration fee of \$35 and a monthly fee of \$25.

Write and solve an equation to determine how many months of membership cost a total of \$210.

Solution

Let

x = the number of months.

The total cost is

$$35 + 25x.$$

Write the equation:

$$35 + 25x = 210.$$

Subtract 35 from both sides:

$$25x = 175.$$

Divide by 25:

$$x = 7.$$

The membership lasts

7

months.

Example 7: Solve the formula

$$A = \frac{1}{2}bh$$

for h .

Solution

Start with

$$A = \frac{1}{2}bh.$$

Multiply both sides by 2:

$$2A = bh.$$

Divide both sides by b :

$$h = \frac{2A}{b}.$$

Summary

Key Ideas

- Word problems can often be solved by writing and solving linear equations.
- Let a variable represent an unknown quantity.
- Write other unknown quantities in terms of that variable when needed.
- Translate the problem conditions into an equation.
- Solve the equation and check the answer in the original wording.
- To solve a formula for a variable, isolate that variable using inverse operations.